Workshop on:
Prompt Engineering, & LLM Fine-tuning on Local Machine for Beginners

Dr. Amit Kumar Mondal,
Associate Professor, CSE KU
amit@cse.ku.ac.bd

Computer Science & Engineering, Khulna University

Contents of This Session

- Basic Understanding of Prompts
- Prompt engineering
- Fine tuning LLM on your local machine
  - Workflow, challenges, usage

Acknowledgement: Resources are collected and prepared with the help of Google search, Gemini, ChatGPT, Medium https://leonnicholls.medium.com/unlock-the-power-of-predictive-ai-with-google-gemini-581e1453d6f2, https://developers.google.com/machine-learning/resources/prompt-eng

GenAI Regular Prompt

What is prompt for GenAI/ChatBot?

- A prompt is the **input (text, image, etc.) you provide to guide the AI to generate a specific response**, acting as instructions, questions, or statements that shape the output's quality, relevance, and format.
- it's how you "talk" to the AI, with clearer, more detailed prompts leading to better, more tailored results.

Usage:

- Creative Writing
- Summarization
- Opinion-Based Questions
- Code completion, Debugging
- Artistic Images
- Strategic Planning

GenAI Regular Prompt

Prompt types:

- Direct prompting (Zero-shot)
- Prompting with examples (One-, few-, and multi-shot)
- Chain-of-thought prompting (CoT)

https://developers.google.com/machine-learning/resources/prompt-eng

Direct prompting (Zero-shot):

Provide:

- Instruction
- Without additional context, without examples

- Idea Generation:

```
Prompt: Can you give me a list of ideas for blog posts for tourists visiting Khulna City for the first time?
```

- Role Prompting:

```
Prompt: You are a mighty and powerful prompt-generating robot. You need to understand my goals and
 objectives and then design a prompt. The prompt should include all the relevant information context and data that was provided to you.  You must continue asking questions  until you are confident that you can produce the best prompt for the best outcome. Your final prompt  must be optimized for chat interactions. Start by asking me to describe my goal, then continue with follow-up questions to design the best prompt.
```

- Data Organization:

```
Prompt: Create a four-column spreadsheet of 10 highly-rated science fiction movies, year of release, average audience rating, and top 3 keywords from audience reviews. Make sure to cite the source of the audience rating.
```

Prompting with examples (One-, few-, and multi-shot):

- providing the model with one or more examples of the desired input-output pairs before presenting the actual prompt

```
Prompt:

Great product, 10/10: Positive
Didn't work very well: Negative
Super helpful, worth it: Positive
It doesn't work!:
```

Chain-of-thought prompting (CoT)

- CoT encourages the model to break down complex reasoning into a series of intermediate steps → more comprehensive and well-structured final output.

```
Prompt:

The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.
A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.
The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.
A:
```

```
Prompt:

I went to the market and bought 10 apples. I gave 2 apples to the neighbor and
2 to the repairman. I then went and bought 5 more apples and ate 1. How many
apples was I left with?

Let's think step by step.
```

https://promptingguide.ai/techniques/cot
https://developers.generativeai.google/prompt-gallery

https://prompthero.com/

Prompting Best Practices

- Clearly communicate what content or information is most important.
- Structure the prompt: Start by defining its role, give context/input data, then provide the instruction.
- Use specific, varied examples to help the model narrow its focus and generate more accurate results.
- Use constraints to limit the scope of the model's output. → avoid factual inaccuracies.
- Break down complex tasks into a sequence of simpler prompts.
- Instruct the model to evaluate or check its own responses before producing them.
- ("Make sure to limit your response to 3 sentences", "Rate your work on a scale of 1-10 for conciseness", "Do you think this is correct?").

https://developers.google.com/machine-learning/resources/prompt-eng

Project: Climate Change and Agricultural Plan for Khulna District

Requirement:

- **Initial Objective:** Create a project for handling agriculture sector in Khulna District for next five years considering the challenges of climate change.
- **Specialty:** Project will focus on grains, area and seasons
- **Provided Data:** previous stat of grains production say 2000-2025, samples doc of previous project

✓ ***Break down complex tasks into a sequence of simpler prompts.***

❑Go to https://tinyurl.com/tbpzavw3

Project: Climate Change and Agricultural Plan for Khulna District

Prompts:

- ***Prompt 1:*** Create a plan for handling agriculture in Khulna District for next five years considering the challenges of climate change.
- ***Prompt 2:*** update the plan for grains such as rice, wheat and dal
- ***Prompt 3:*** update the plan for season wise and area wise for khulna distric
- ***Prompt 4:*** include Gantt chart, probable cost, and manpower needed area wise
- ***Prompt 5:*** Modify and merge the doc that Includes grains production data and bart chart for it for 2000-2025 before section 2. Season-Wise Crop Calendar (2026)
- ***Prompt 6:*** Include useful and low cost AI tools for efficient implementation of the plan

Prompt Engineering for Strategic Planning

Regular vs Causal Prompt

- Regular prompt – information like query
- Example: “What are the benefits of electric cars?”
- “Causal prompts,”– “what if” scenario
- causal prompts encourage GenAI to analyze cause-and-effect relationships, predict potential outcomes, and explore different scenarios
- Example -- “What would happen if electric cars became the dominant mode of transportation?”
- Gemini has to think about the chain reactions — the impact on oil prices, the demand for charging stations, the changes in urban planning, and so on.

Key Strategic Planning Steps

1.Establish Foundation & Vision:

- Define government vision, mission, and core goals (e.g., save lives, protect investments).
- Create a planning team and define roles.
- Develop overarching policy and legal frameworks (Acts, Policies).

2.Risk Assessment & Analysis:

- Conduct comprehensive assessments of potential hazards (floods, cyclones, etc.) and vulnerabilities.
- Analyze impacts on various sectors (social, economic, infrastructure).

3.Prevention & Mitigation:

- Implement policies and investments to reduce disaster risk *before* an event.
- Build resilient infrastructure and incorporate disaster risk reduction (DRR) into development.

4.Preparedness:

- Develop hazard-specific action plans and standing orders.
- Invest in training, exercises, and reliable communication systems.
- Strengthen resource management systems and ensure social protection.

Key Strategic Planning Steps

5.Response:

- Activate plans for immediate actions during a disaster (warning, search & rescue, emergency relief).
- Coordinate across agencies and ensure clear communication.

6.Recovery & Rehabilitation:

- Plan for post-disaster reconstruction, rehabilitation, and long-term resilience building.
- Integrate lessons learned into future planning.

7.Governance & Continuous Improvement:

- Ensure strong institutional structures and accountability.
- Conduct after-action reviews, annual reviews, and capture lessons learned for policy development and continuous improvement.
- Promote collaboration and integrated planning across all government levels and sectors.

Structure 1: Single Future Scenario Analysis

- Exploring the potential consequences of a specific event or trend

Analyze ***[future scenario]*** by:

- 1. **Initial Conditions**: List key factors influencing this scenario.
- 2. **Trigger Events:** Identify potential catalysts to set this future in motion.
- 3. **Causal Chain:** Map out the sequence of:
  - * Primary effects (0-6 months)
  - * Secondary effects (6-18 months)
  - * Tertiary effects (18+ months)
- 4. **Decision Points:** Identify critical moments where different choices could alter outcomes.
- 5. **Leading Indicators:** List early warning signs that this scenario is more likely.

Structure your response as an ***if-then-therefore*** chain of events.

Prompt 1: Single Future Scenario Analysis

- **Example Prompt:**
  "Analyze a future where virtual reality becomes how people interact with each other and the world."

- **Real-World Application:** This template is excellent for:
- **Forecasting technological disruptions:** "What happens if AI can write better code than humans?"
- **Anticipating social changes:** "What if remote work becomes the norm?"
- **Exploring environmental challenges:** "What if a major city runs out of water?"

Prompt 2: Multiple Scenario Comparison

➢"compare and contrast" with different potential futures-- probabilities, identify key decision points, and more informed choices

➢ For ***[project/situation],*** analyze three possible futures:

* Baseline Scenario (Most Likely)
* Optimistic Scenario
* Challenge Scenario

➢ **For each scenario**:

* Identify three key triggering events.
* Map causal relationships between events.
* Describe feedback loops and amplifying factors.
* List early warning indicators.
* Specify decision points for intervention.

➢ **Compare scenarios based on:**

* Probability of critical triggers.
* Speed of cause-effect chains.
* Potential for intervention.
* Critical dependencies.

Prompt 2: Multiple Scenario Comparison

- **Example Prompt:**

"For the future of space exploration, analyze three possible scenarios."

- **Real-World Application:** This template is invaluable for:
- **Strategic planning:** "What are the different ways our business could evolve over the next 5 years?"
- **Decision-making under uncertainty:** "What are the potential outcomes of this investment, and how can we prepare for each?"
- **Creative brainstorming:** "What are three different ways this story could end?"

Prompt 3: Strategic Planning

- combines future scenario analysis and risk assessment elements → for a comprehensive plan for achieving a specific goal.

Develop a strategic plan for ***[objective]*** by:

➢ **1. Current State Analysis:**

* List 3 existing trends supporting the objective.

* Identify 2 potential barriers.

* Map current resources and capabilities.

➢ **2. Future State Mapping:**

* Define 3 sequential milestones.

* For each milestone:

    * Required preconditions.

    * Causal triggers needed.

    * Expected outcomes.

    * Potential complications.

➢ **3. Intervention Points:**

* Identify 3 critical decision moments.

* For each decision:

    * Required inputs.

    * Potential outcomes.

    * Cascading effects.

    * Risk mitigation options.

➢ **4. Success Indicators:**

* Early warning signs of success/failure.

* Measurable intermediate outcomes.

* Long-term success metrics.

Prompt 3: Strategic Planning

Example Prompt: *"Develop a strategic plan for reducing carbon emissions in a major city by 50% within the next 10 years."*

- **Real-World Application:** This template is incredibly versatile and can be used for:
- **Business strategy:** "How can we increase market share by 20% next year?"
- **Project management:** "What are the key steps in completing this project on time and within budget?"
- **Personal goal setting:** "How can I achieve my goal of running a marathon in the next 6 months?"

Pro Tip: Try to tackle only some things at a time. Break down complex plans into smaller, more manageable steps, and use Gemini to analyze each step and identify potential roadblocks.

Refining Prompt if You are Stuck

- Repeat key words, phrases, or ideas
- Specify your desired output format (CSV, JSON, etc.)
- Use all caps to stress important points or instructions. You can also try exaggerations or hyperbolic language; for example: "Your explanation should be absolutely impossible to misinterpret. Every single word must ooze clarity!"
- Use synonyms or alternate phrasing (e.g., instead of "Summarize," try appending "tldr" to some input text). Swap in different words or phrases and document which ones work better and which are worse.
- Try the sandwich technique with long prompts: Add the same statement in different places.
- Use a prompt library for inspiration.

https://prompthero.com/

https://developers.generativeai.google/prompt-gallery

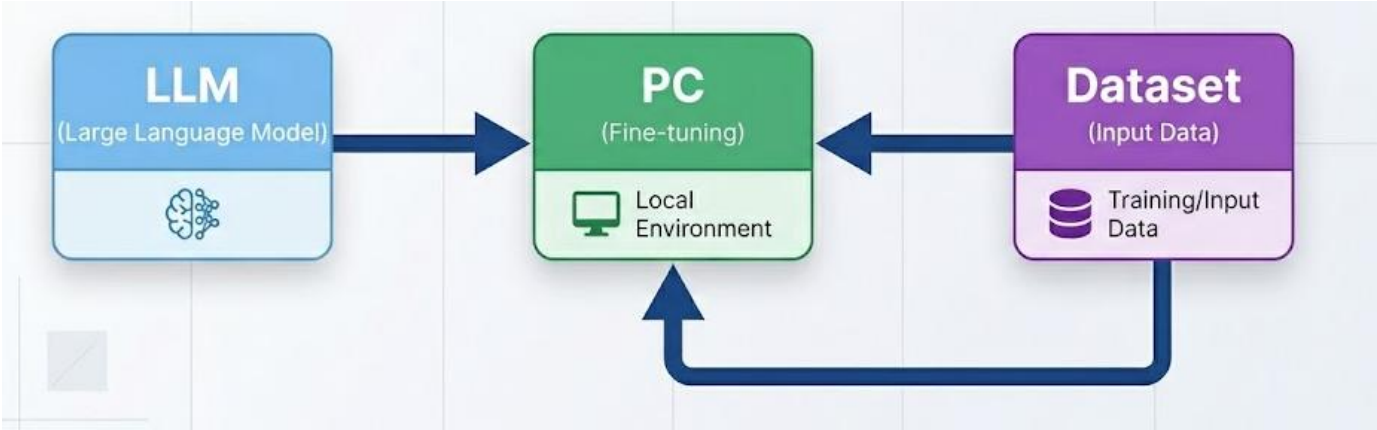
https://developers.google.com/machine-learning/resources/prompt-eng

Practice project: Predicting and Strategic Planning for managing Extreme cold weather (30 days) impacts for Khulna city Considering the poor and homeless people

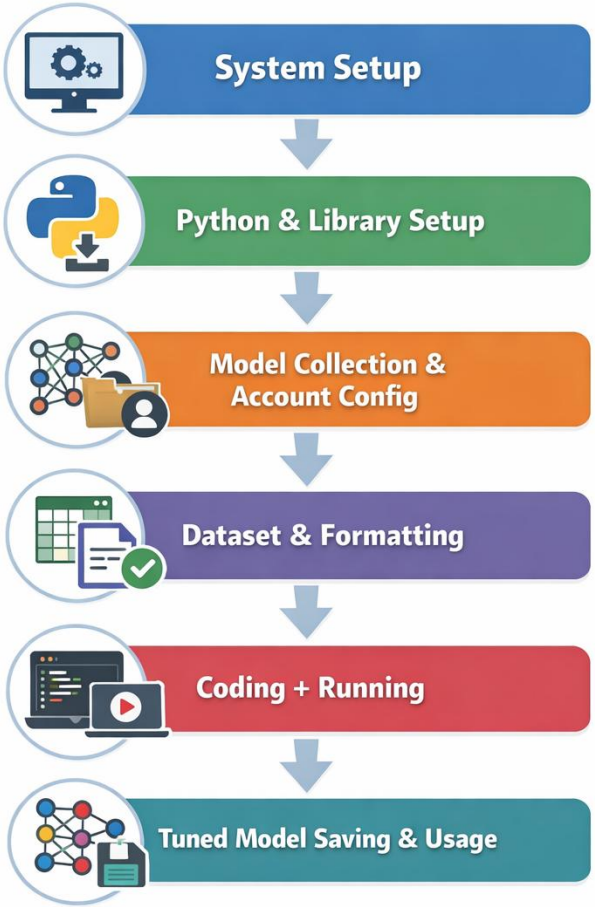
Example Prompt:

Pro Tip: Try to tackle only some things at a time. Break down complex plans into smaller, more manageable steps, and use Gemini to analyze each step and identify potential roadblocks.

Fine Tuning Gemma3 in Local Machine



Fine Tuning Workflow



System Setup

- Remember: **tensorflow-text** does not work in Windows, so do not try in bare windows
- For windows: install ***wsl*** and ***ubuntu*** from ***powershell***
  - wsl --install -d Ubuntu
- Configure windows (I got proper help from Gemini):
- Search for **"Turn Windows features on or off"** in the Start menu.
- Ensure these three are **checked**:
  - **Virtual Machine Platform**
  - **Windows Subsystem for Linux**
  - **Hyper-V**.
- Click OK and **Restart**.

Python and library setup

- Open **wsl** in admin mode. Setup account
- Need at least **python 3.10**
- Setup repo for python 3.10 installation
  - sudo apt update && sudo apt upgrade -y
  - sudo apt install software-properties-common -y
  - sudo add-apt-repository ppa:deadsnakes/ppa sudo apt update
  - sudo apt install python3.10 python3.10-venv python3.10-dev -y
  - mkdir ~/gemmaproject && cd ~/gemma_project

- – Create python virtual environment
  - python3.10 -m venv gemma_env

  Run virtual environment
  - source gemma_env/bin/activate

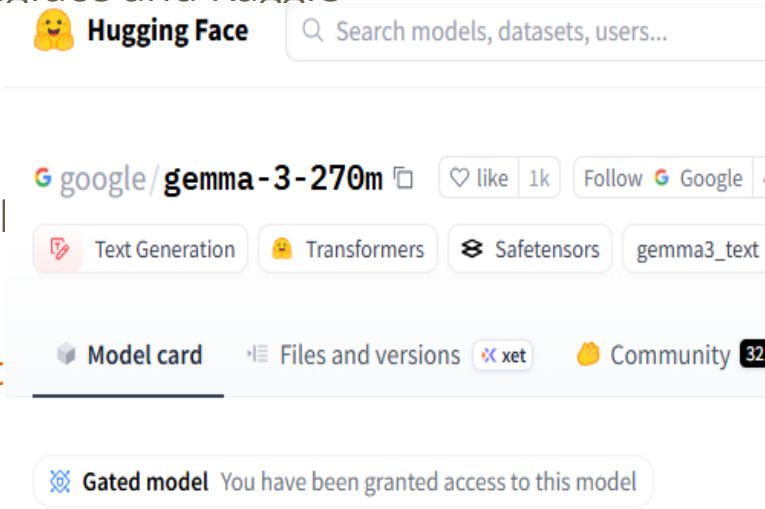
Python and library setup

frozenlist	1.8.0
fsspec	2026.2.0
gast	0.7.0
gguf	0.17.1
google-pasta	0.2.0
greenlet	3.3.2
grpcio	1.78.1
h11	0.16.0
h5py	3.15.1
hf-xet	1.2.0
httpcore	1.0.9
httpx	0.28.1
huggingface_hub	0.36.2
idna	3.11
iniconfig	2.3.0
jiter	0.13.0
kagglehub	1.0.0
kagglesdk	0.1.15
keras	3.12.1
keras-hub	0.25.1

sniffio	1.3.1
sympy	1.13.1
tensorboard	2.20.0
tensorboard-data-server	0.7.2
tensorflow	2.20.0
tensorflow-text	2.20.0
termcolor	3.3.0
tokenizers	0.22.2
tomli	2.4.0
torch	2.6.0+cp
tqdm	4.67.3
transformers	4.57.6
typer	0.15.4
typer-slim	0.24.0
typing-inspection	0.4.2
typing_extensions	4.15.0
tzdata	2025.3
urllib3	2.6.3

Model Collection & Account Config

- various opensource models are available at Huggingface and Kaggle
- – Hugging Face
- Create account into Hugging Face,
- Generate a token having at least read access. Use tl loading code
- Then configure the gemma3 for licensing authentic




```
os.environ["KERAS_BACKEND"] = "tensorflow"
os.environ["HF_TOKEN"] = "hf_.....SA"
model_id = "hf://google/gemma-3-270m"
gemma_lm = keras_hub.models.Gemma3CausalLM.from_preset(model_id)
```

Dataset & Formatting

- Example dataset: martian dataset
  - https://huggingface.co/datasets/bebechien/MobileGameNPC/resolve/main/martian.csv

```
player: "Who are you?", alien: "I am Zorp. Juzt a humble rock-drinker
from da Great Red-Dust. You... you z-mell like... wet plantz."
```

- Code:

```
training_data = {
    "prompts": [row['player'] for row in all_rows],
    "responses": [row['alien'] for row in all_rows]
}
```

Coding + running

- – Install pycharm through wsl
- Run: cd pycharm-community-*/bin ./pycharm.sh
- **Sample code is available here:** https://medium.com/@abhishek20dgp/building-a-bangla-llm-by-finetuning-gemma-2b-llm-using-low-rank-adaptation-lora-73bad579f0a4
- – Code using pycharm, and use the created virtual environment within the project
- – Run the code from wsl pycharm
- – Its better to run the code from command line (I was successful)
    - python3.10 -m model.train_martian

Algorithm Configuration

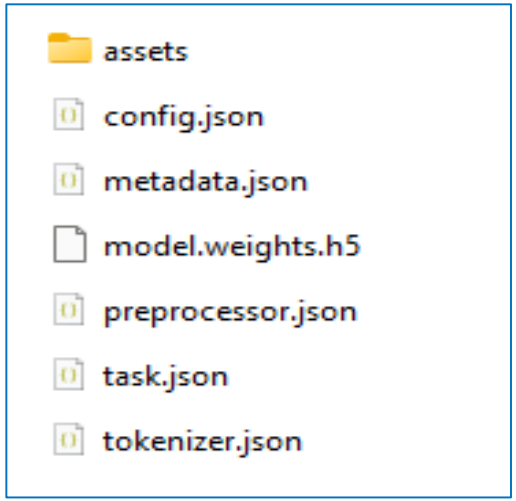
- LOCUS: LORA
- Epoch size
- Training Test split
- Accuracy Tracking

```
gemma_lm.backbone.enable_lora(rank=16)

formatted_train_data = get_prompts_format()

gemma_lm.compile(
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    optimizer=keras.optimizers.Adam(learning_rate=5e-5),
    weighted_metrics=["accuracy"]
     )
gemma_lm.fit(formatted_train_data, epochs=22, batch_size=1)
```

Model Saving

- **Constraints for saving:**
- Memory
- Model size
- Inconsistent state

Pre-trained model structure




```
gc.collect()
try:
    gemma_lm.save_to_preset("gemma3_cot_local.keras")
except Exception:
    logging.error("An error occurred during execution",
exc_info=True)
if hasattr(os, 'sync'):
    os.sync()
```

Tuned Model Loading & Usage

```
def test_martian_npc(user_input):
    prompt = (
        f"<|role_start|>user<|role_metadata|>\n{user_input}<|role_end|>\n"
        f"<|role_start|>assistant<|role_metadata|>\n"
    )

    output = gemma_lm.generate(prompt, max_length=128)

    response = output.split("<|role_start|>assistant<|role_metadata|>\n")[-1]
    response = response.replace("<|role_end|>", "").strip()
```

“Software is Eating the World”

“Data is Gold”

----------“AI is Power”----------

Thank you!

Contact: amit@cse.ku.ac.bd

https://ku.ac.bd/discipline/cse/faculty/amit